

Übung 7 – Dioden

Aufgabe 1: Diode als Schalter: Einphasiger Brückengleichrichter

Vier Dioden seien als Brückengleichrichter zusammen mit einer Wechselspannungsquelle $u_i = 325 \text{ V} \sin(\omega t)$ angeordnet, wie in Abbildung 1 gezeigt.

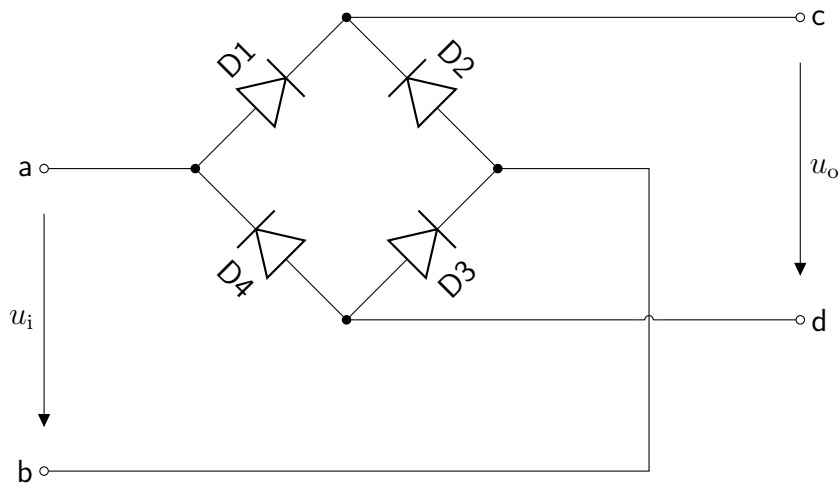
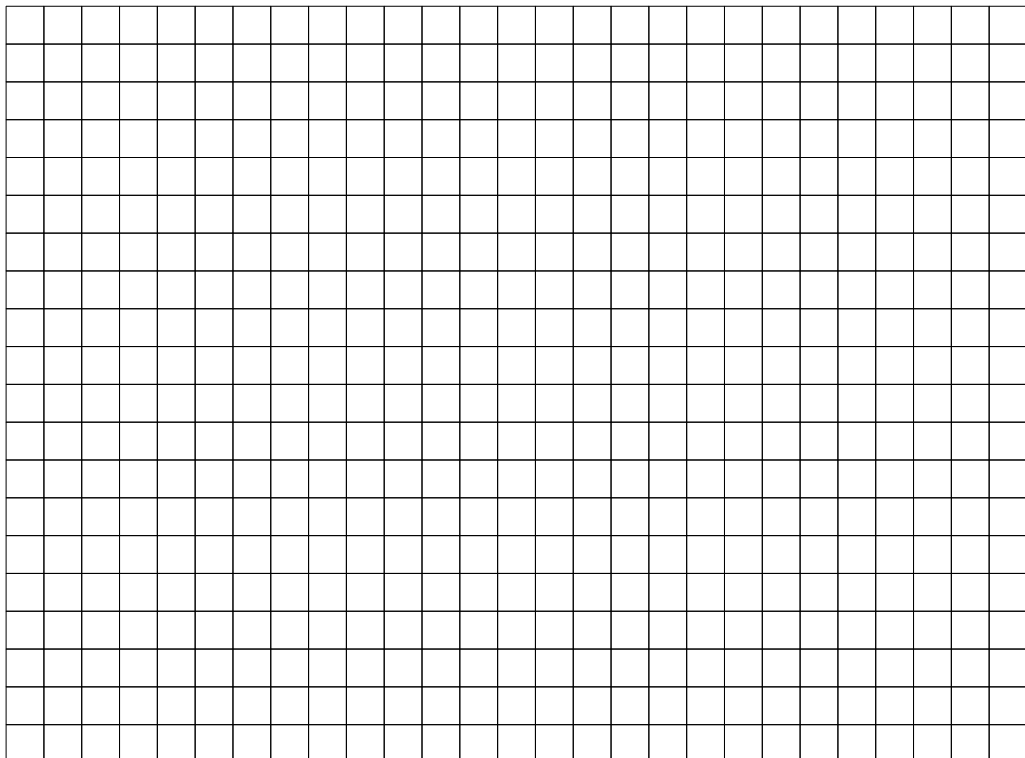


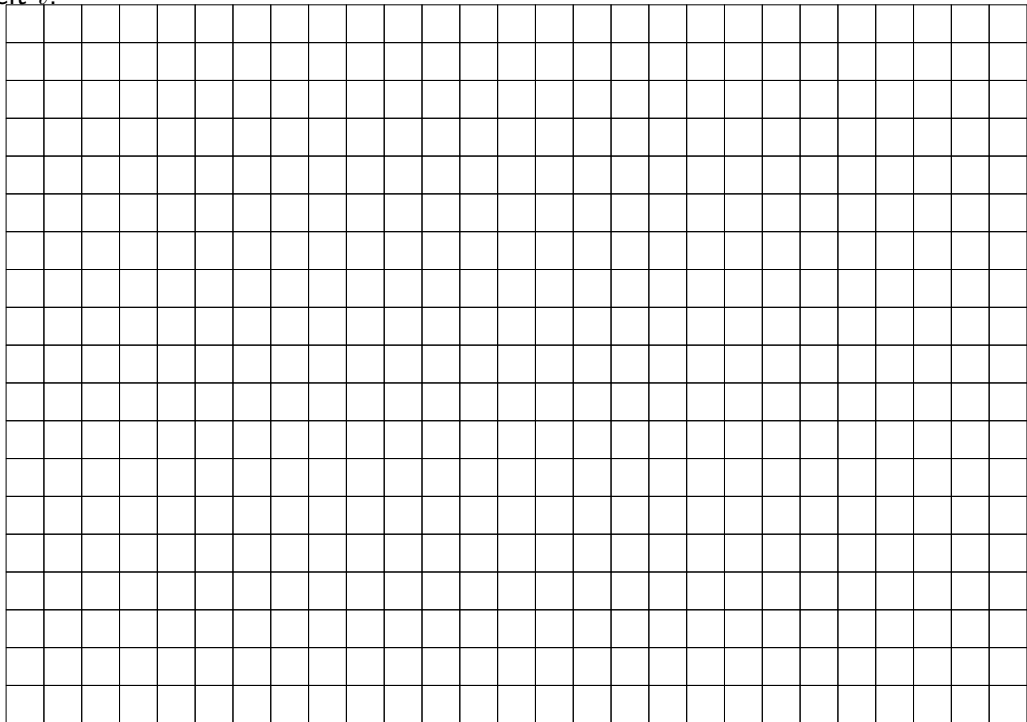
Abbildung 1: Ein einphasiger Brückengleichrichter.

- a) Modellieren Sie die Dioden als Schalter, indem Sie die Zustände „offen“ oder „geschlossen“ in Bezug auf die Eingangsspannung betrachten. Unterscheiden Sie zwischen den Fällen „a auf hohem Potenzial, b auf niedrigem Potenzial“ und „a auf niedrigem Potenzial, b auf hohem Potenzial“.

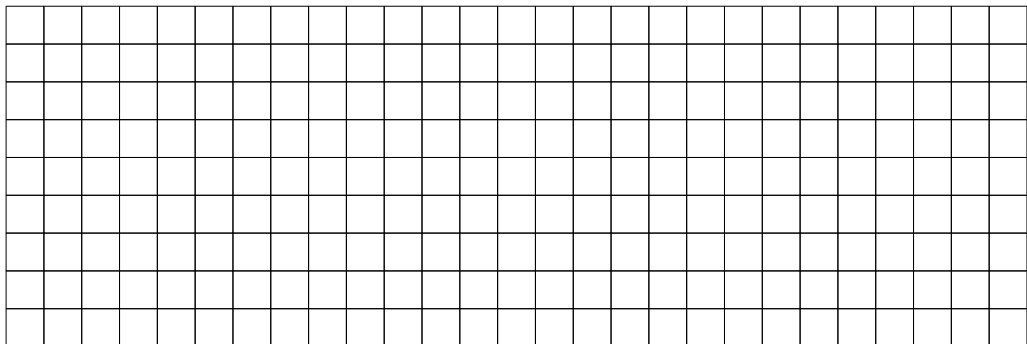
Tipp: Markieren Sie die Knoten mit gleichem Potenzial.



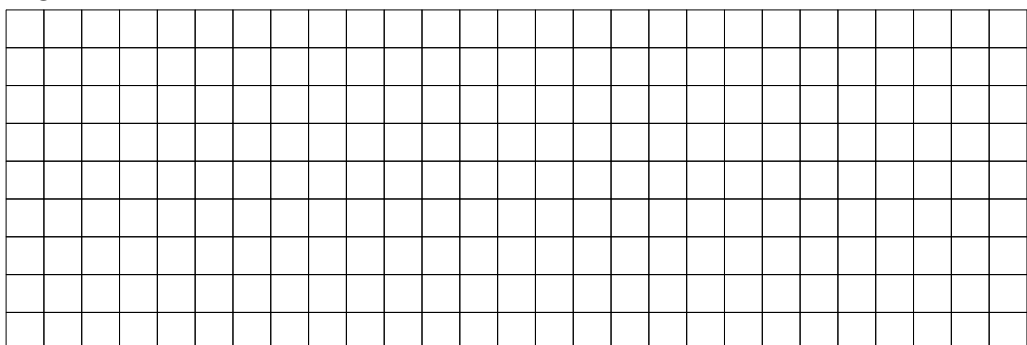
- b) Bitte zeichnen Sie die Eingangsspannung u_i und die Ausgangsspannung u_o über die Zeit t .



- c) Berechnen Sie die effektive Ausgangsspannung U_o und vergleichen Sie sie mit der Eingangsspannung U_i .



- d) Mit einem einzigen Bauteil können Sie das Ausgangssignal glätten. Um welches Bauteil handelt es sich? Bitte zeichnen Sie es in das Schaltbild ein und fügen Sie den neuen Verlauf der Ausgangsspannung u_o in Ihre Darstellung oben ein. Was stellen Sie bezüglich des Effektivwertes fest?



Aufgabe 2: Diode als Konstantspannungsquelle: Spannungsstabilisierung

Eine Z-Diode mit $U_Z = 6,8 \text{ V}$ und $r_D = 0 \Omega$ sei in Serie mit einem Widerstand $R_s = 100 \Omega$ an eine dreieckförmige Eingangsspannung $5 \text{ V} \leq u_i \leq 10 \text{ V}$ angeschlossen.

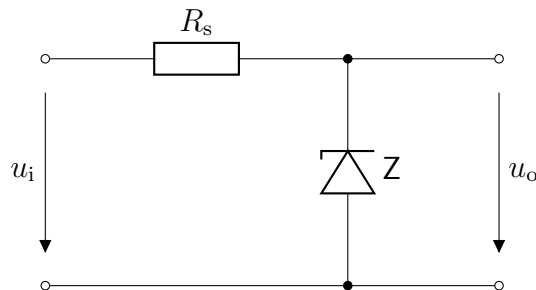
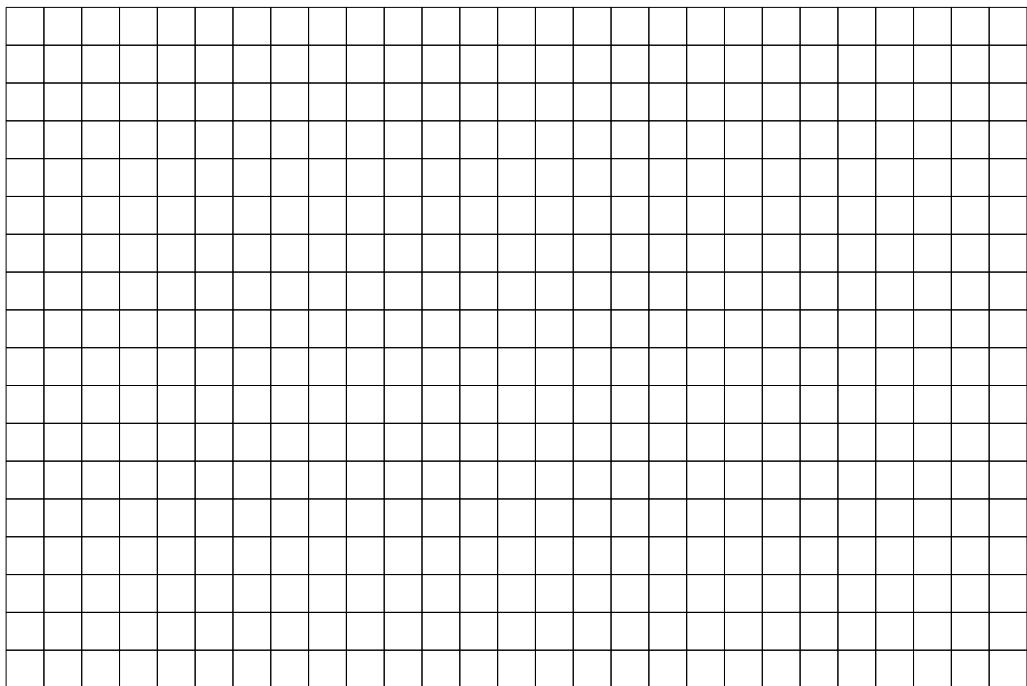
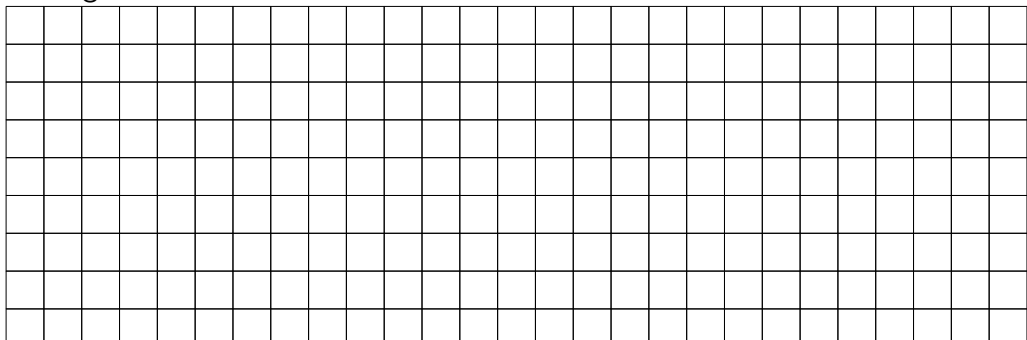


Abbildung 2: Eine Zenerdiode in einer Spannungsstabilisierungsschaltung.

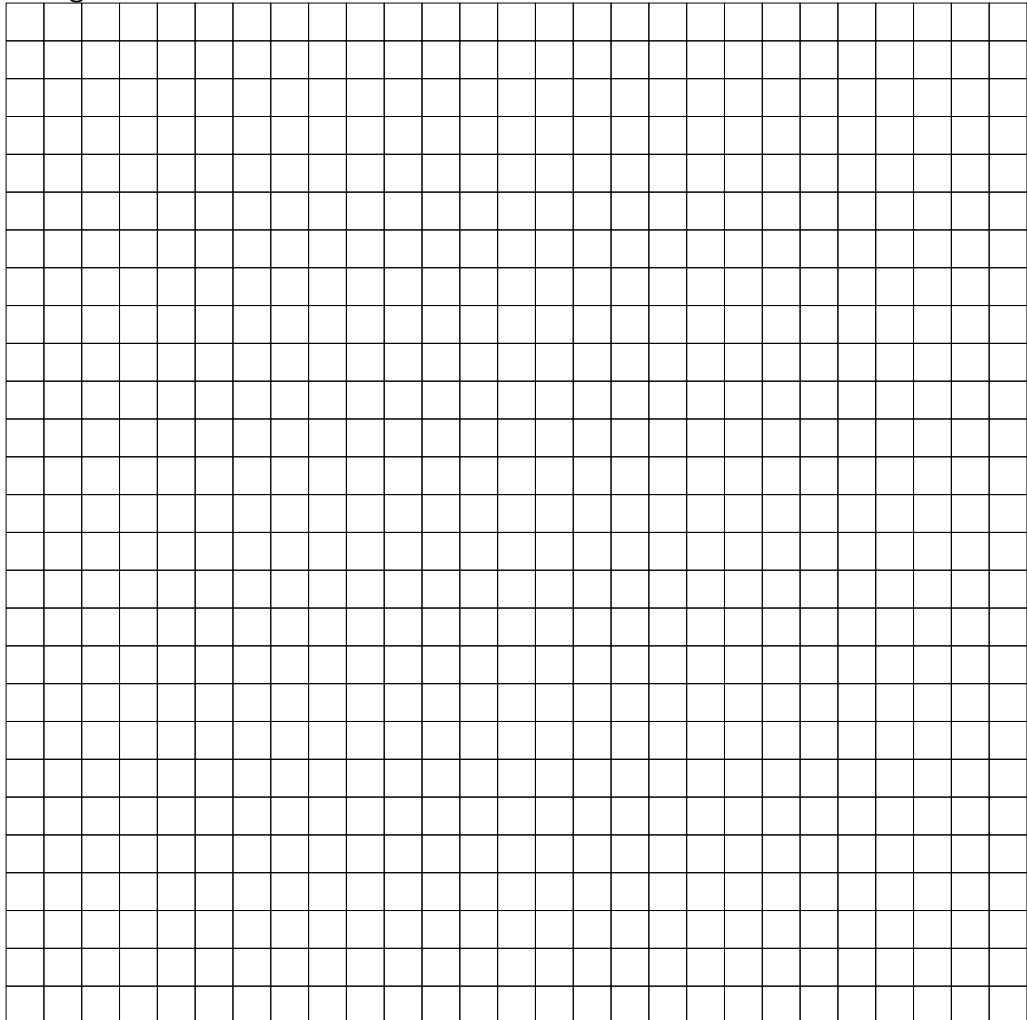
- a) Bitte zeichnen Sie ein Schaltbild, das die Diode durch einen Schalter und eine konstante Spannungsquelle ersetzt.



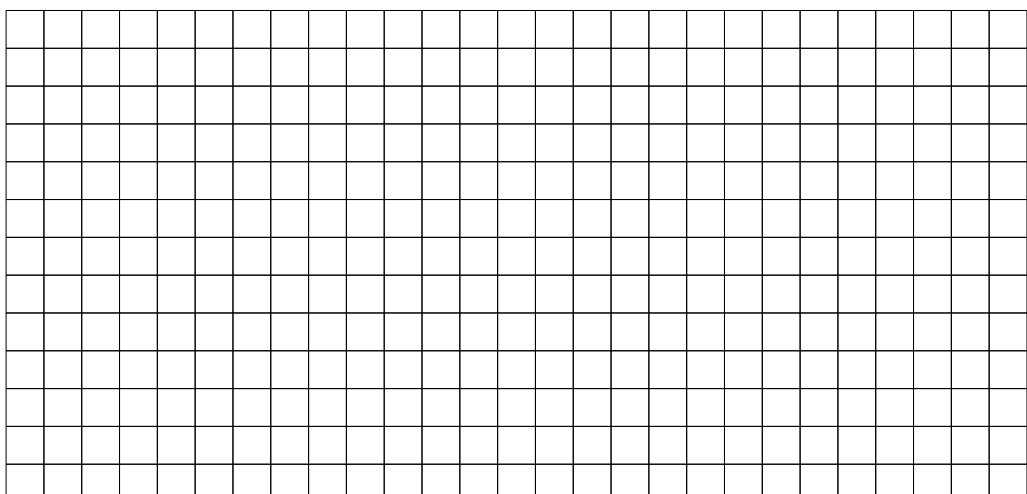
- b) Beschreiben Sie die beiden Betriebsmodi dieses Modells in Bezug auf die Eingangsspannung u_i .



- c) Berechnen Sie den maximalen Diodenstrom $I_{Z,\max}$ und die maximale Leistung $P_{Z,\max}$, die von der Diode unter Leerlaufbedingungen verbraucht wird. Überprüfen Sie das beigefügte Datenblatt, ob dies innerhalb der Betriebsgrenzen der Diode liegt. Welche Rolle spielt der Lastwiderstand im Allgemeinen bei der von der Diode verbrauchten Leistung?



- d) Zeichnen Sie die Eingangs- und Ausgangsspannung als Funktion der Zeit.



Aufgabe 3: Graphische Ermittlung des Arbeitspunkts

In dieser Aufgabe wird das Verhalten einer idealen Diode ($r_D = 0$) aus Aufgabe 2 mit einer realen Z-Diode (Zetex BZX84C6V8, $r_D \neq 0$) verglichen. Der Arbeitspunkt der Diode soll anhand der Kennlinie in Bild 3 und der für den Schaltkreis 2 aus Aufgabe 2 ermittelten Arbeitsgerade bestimmt werden.

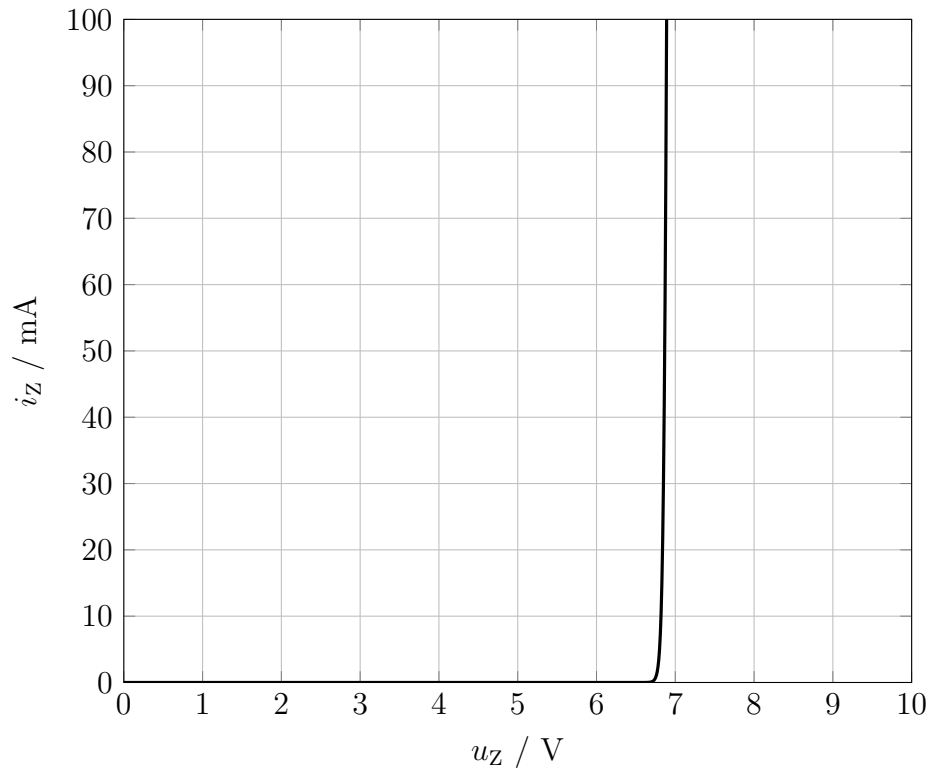
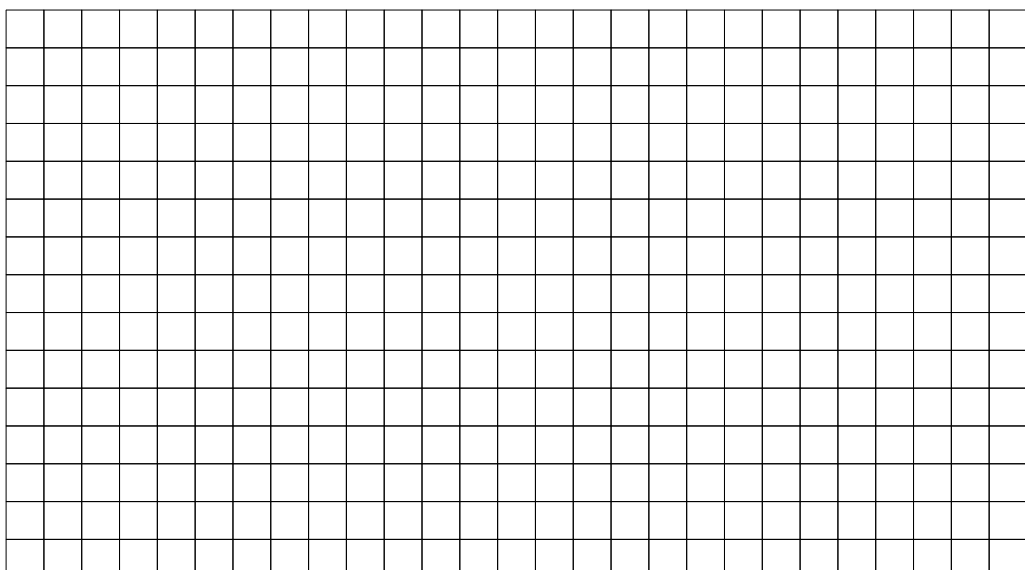
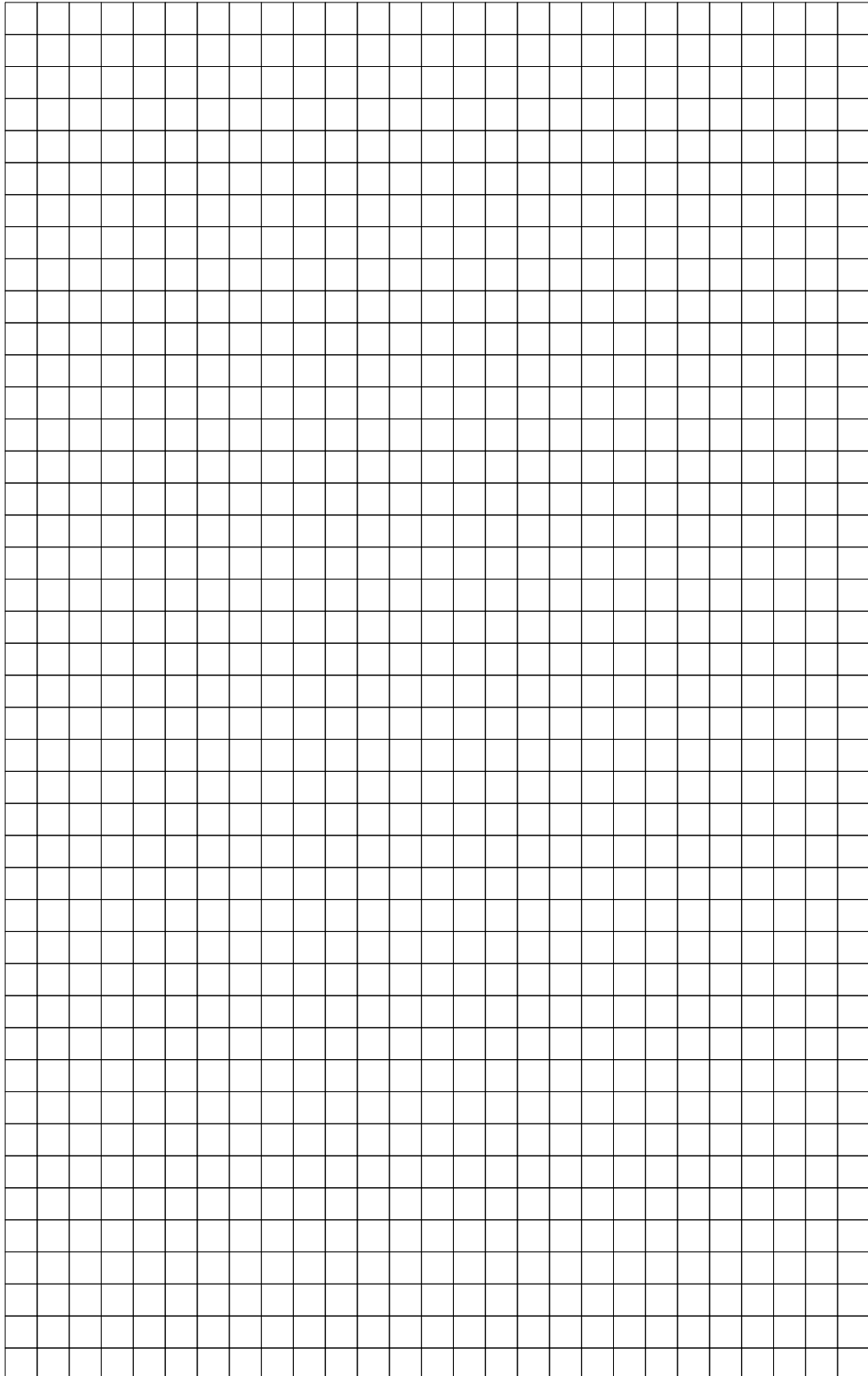


Abbildung 3: Kennlinie einer 6,8 V Zener-Diode (Zetex BZX84C6V8).

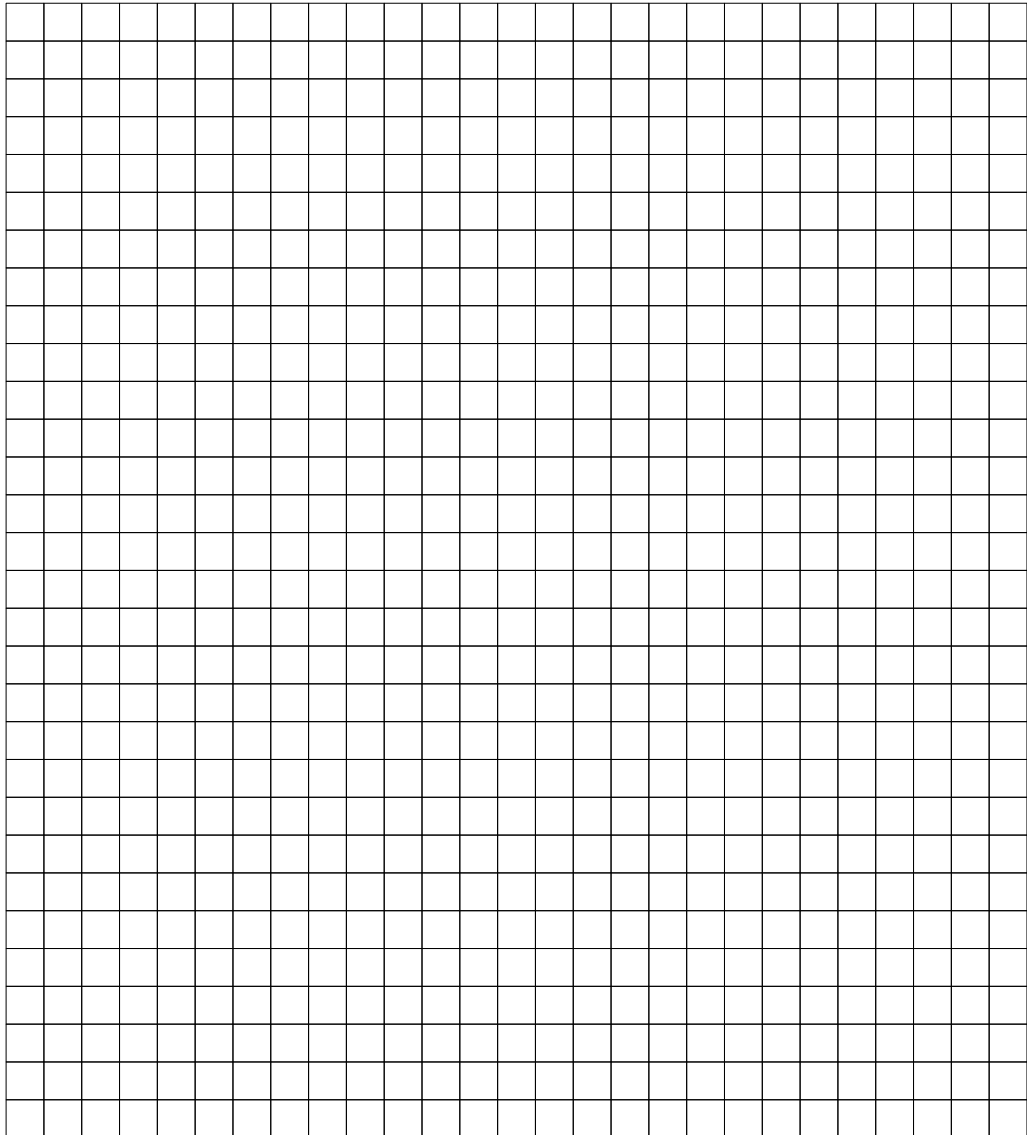
- a) Zeichnen Sie die Arbeitsgerade und bestimmen Sie die Arbeitspunkte $Q(U_Z, I_Z)$ für drei verschiedene Eingangsspannungen $U_{i,1} = 5 \text{ V}$, $U_{i,2} = 7,5 \text{ V}$ und $U_{i,3} = 10 \text{ V}$.



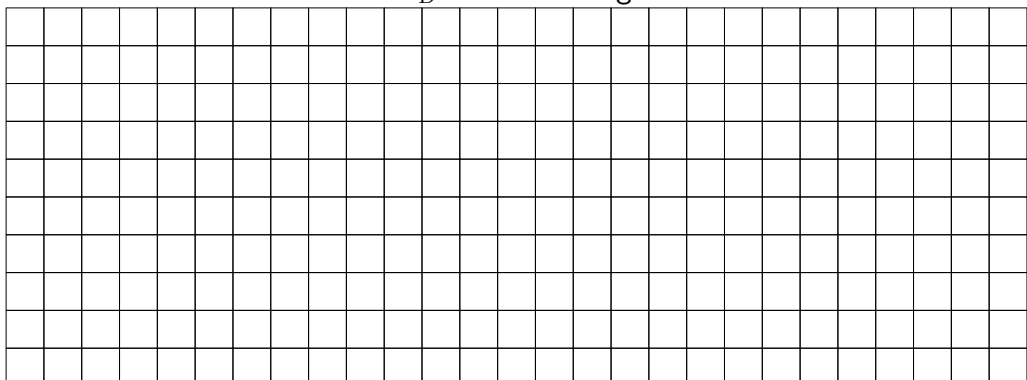
- b) Für die meisten Dioden ist ein minimaler Zenerprüfstrom I_{ZT} für den stabilen Betrieb der Diode vom Hersteller angegeben. Finden Sie den minimalen Nennstrom im beigefügten Datenblatt. Was können Sie über die Arbeitspunkte Q_1 , Q_2 und Q_3 aussagen? Vergleichen Sie das Ergebnis mit Aufgabe 2.



- b) Stellen Sie die Übertragungsfunktion u_o als Funktion von u_i im Bereich $-5\text{ V} \leq u_i \leq 5\text{ V}$ grafisch dar. Stellen Sie dann das Ausgangssignal für zwei unterschiedliche Eingangssignale $u_{i,1} = 2\text{ V} \sin(\omega t)$ und $u_{i,2} = 4\text{ V} \sin(\omega t)$ dar, indem Sie die Eingangssignale an der Übertragungsfunktion spiegeln.



- c) Wie würde sich die Form der Übertragungskurve qualitativ ändern, wenn der differentielle Innenwiderstand der Diode $r_D = 10\Omega$ betragen würde?



Features

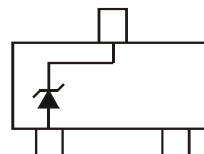
- Planar Die Construction
- 350mW Power Dissipation
- Zener Voltages from 2.4V - 39V
- Ideally Suited for Automated Assembly Processes
- Lead, Halogen and Antimony Free, RoHS Compliant "Green" Device (Notes 3 and 4)**
- Qualified to AEC-Q101 Standards for High Reliability**

Mechanical Data

- Case: SOT-23
- Case Material: Molded Plastic. UL Flammability Classification Rating 94V-0
- Moisture Sensitivity: Level 1 per J-STD-020
- Terminals: Matte Tin Finish annealed over Alloy 42 leadframe (Lead Free Plating). Solderable per MIL-STD-202, Method 208
- Polarity: See Diagram
- Marking Information: See Page 3
- Ordering Information: See Page 3
- Weight: 0.008 grams (approximate)



Top View



Device Schematic

Maximum Ratings @T_A = 25°C unless otherwise specified

Characteristic	Symbol	Value	Unit
Forward Voltage @ I _F = 10mA	V _F	0.9	V

Thermal Characteristics

Characteristic	Symbol	Value	Unit
Power Dissipation (Note 1)	P _D	300	mW
Power Dissipation (Note 2)	P _D	350	mW
Thermal Resistance, Junction to Ambient Air (Note 1)	R _{θJA}	417	°C/W
Thermal Resistance, Junction to Ambient Air (Note 2)	R _{θJA}	357	°C/W
Operating and Storage Temperature Range	T _J , T _{STG}	-65 to +150	°C

- Notes:
- Device mounted on FR-4 PC board with recommended pad layout, which can be found on our website at <http://www.diodes.com/datasheets/ap02001.pdf>.
 - Valid provided the terminals are kept at ambient temperature.
 - No purposefully added lead. Halogen and Antimony Free.
 - Product manufactured with Data Code OW (week 42, 2009) and newer are built with Green Molding Compound. Product manufactured prior to Date Code OW are built with Non-Green Molding Compound and may contain Halogens or Sb₂O₃ Fire Retardants.

Electrical Characteristics @T_A = 25°C unless otherwise specified

Type Number	Marking Code	Zener Voltage Range (Note 5)				Maximum Zener Impedance (Note 6)			Maximum Reverse Current (Note 5)		Typical Temperature Coefficient @ I _{ZT} mV/°C	
		V _Z @ I _{ZT}			I _{ZT} (mA)	Z _{KT} @ I _{ZT} (Ω)	Z _{ZK} @ I _{ZK}		I _R (μA)	V _R (V)	Min	Max
		Nom (V)	Min (V)	Max (V)			(Ω)	(mA)				
BZX84C2V4	KZB	2.4	2.2	2.6	5.0	100	600	1.0	50	1.0	-3.5	0
BZX84C2V7	KZC	2.7	2.5	2.9	5.0	100	600	1.0	20	1.0	-3.5	0
BZX84C3V0	KZD	3.0	2.8	3.2	5.0	95	600	1.0	10	1.0	-3.5	0
BZX84C3V3	KZE	3.3	3.1	3.5	5.0	95	600	1.0	5.0	1.0	-3.5	0
BZX84C3V6	KZF	3.6	3.4	3.8	5.0	90	600	1.0	5.0	1.0	-3.5	0
BZX84C3V9	KZG	3.9	3.7	4.1	5.0	90	600	1.0	3.0	1.0	-3.5	0
BZX84C4V3	KZH	4.3	4.0	4.6	5.0	90	600	1.0	3.0	1.0	-3.5	0
BZX84C4V7	KZ1	4.7	4.4	5.0	5.0	80	500	1.0	3.0	2.0	-3.5	0.2
BZX84C5V1	KZ2	5.1	4.8	5.4	5.0	60	480	1.0	2.0	2.0	-2.7	1.2
BZX84C5V6	KZ3	5.6	5.2	6.0	5.0	40	400	1.0	1.0	2.0	-2.0	2.5
BZX84C6V2	KZ4	6.2	5.8	6.6	5.0	10	150	1.0	3.0	4.0	0.4	3.7
BZX84C6V8	KZ5	6.8	6.4	7.2	5.0	15	80	1.0	2.0	4.0	1.2	4.5
BZX84C7V5	KZ6	7.5	7.0	7.9	5.0	15	80	1.0	1.0	5.0	2.5	5.3
BZX84C8V2	KZ7	8.2	7.7	8.7	5.0	15	80	1.0	0.7	5.0	3.2	6.2
BZX84C9V1	KZ8	9.1	8.5	9.6	5.0	15	100	1.0	0.5	6.0	3.8	7.0
BZX84C10	KZ9	10	9.4	10.6	5.0	20	150	1.0	0.2	7.0	4.5	8.0
BZX84C11	KY1	11	10.4	11.6	5.0	20	150	1.0	0.1	8.0	5.4	9.0
BZX84C12	KY2	12	11.4	12.7	5.0	25	150	1.0	0.1	8.0	6.0	10.0
BZX84C13	KY3	13	12.4	14.1	5.0	30	170	1.0	0.1	8.0	7.0	11.0
BZX84C15	KY4	15	13.8	15.6	5.0	30	200	1.0	0.1	10.5	9.2	13.0
BZX84C16	KY5	16	15.3	17.1	5.0	40	200	1.0	0.1	11.2	10.4	14.0
BZX84C18	KY6	18	16.8	19.1	5.0	45	225	1.0	0.1	12.6	12.4	16.0
BZX84C20	KY7	20	18.8	21.2	5.0	55	225	1.0	0.1	14.0	14.4	18.0
BZX84C22	KY8	22	20.8	23.3	5.0	55	250	1.0	0.1	15.4	16.4	20.0
BZX84C24	KY9	24	22.8	25.6	5.0	70	250	1.0	0.1	16.8	18.4	22.0
BZX84C27	KYA	27	25.1	28.9	2.0	80	300	0.5	0.1	18.9	21.4	25.3
BZX84C30	KYB	30	28.0	32.0	2.0	80	300	0.5	0.1	21.0	24.4	29.4
BZX84C33	KYC	33	31.0	35.0	2.0	80	325	0.5	0.1	23.1	27.4	33.4
BZX84C36	KYD	36	34.0	38.0	2.0	90	350	0.5	0.1	25.2	30.4	37.4
BZX84C39	KYE	39	37.0	41.0	2.0	130	350	0.5	0.1	27.3	33.4	41.2

Notes: 5. Short duration pulse test used to minimize self-heating effect.
6. f = 1KHz.

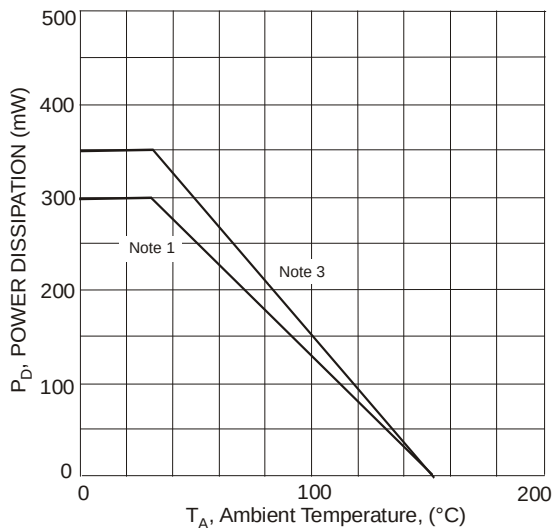


Fig. 1 Power Derating Curve

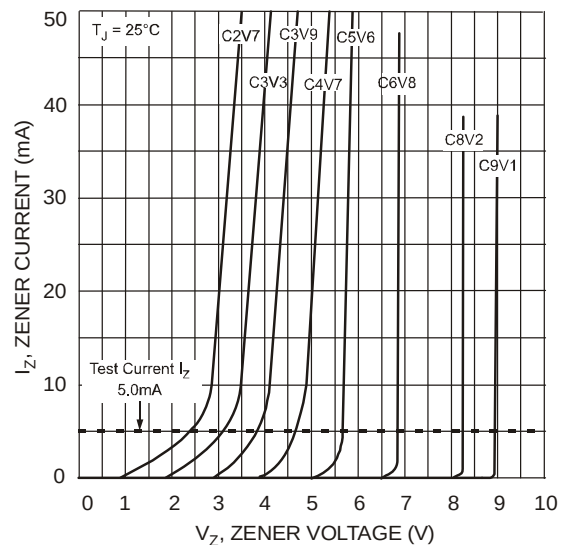


Fig. 2 Typical Zener Breakdown Characteristics

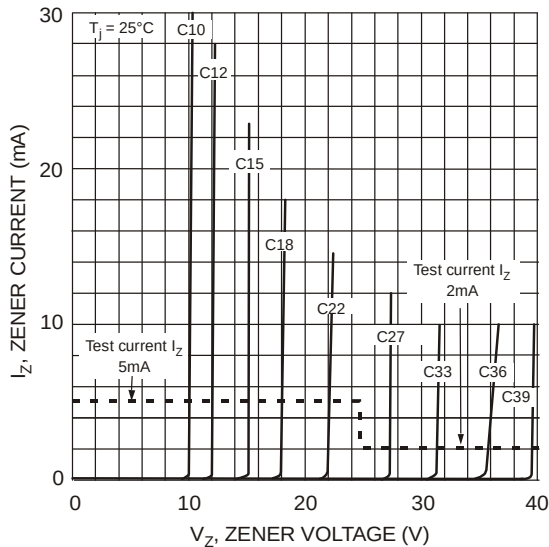


Fig. 3 Typical Zener Breakdown Characteristics

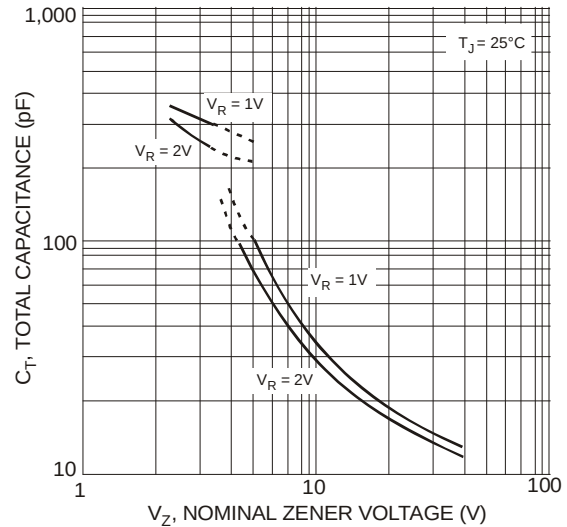


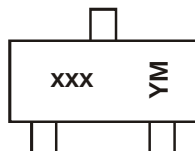
Fig. 4 Typical Total Capacitance vs. Nominal Zener Voltage

Ordering Information (Note 7)

Part Number	Case	Packaging
(Type Number)-7-F (Note 8)	SOT-23	3,000/Tape & Reel
(Type Number)-13-F (Note 9)	SOT-23	10,000/Tape & Reel

- Notes:
- For packaging details, go to our website at <http://www.diodes.com/datasheets/ap02007.pdf>.
 - Add "-7-F" to the appropriate type number in Electrical Characteristics Table on Page 2. Example: 6.2V Zener = BZX84C6V2-7-F.
 - Add "-13-F" to the appropriate type number in Electrical Characteristics Table on Page 2. Example: 2.7V Zener = BZX84C2V7-13-F. Please note: Not all voltages are available in 13" reel size. Please contact the Diodes Inc. Sales Department for assistance in ordering 13" reel size devices.

Marking Information



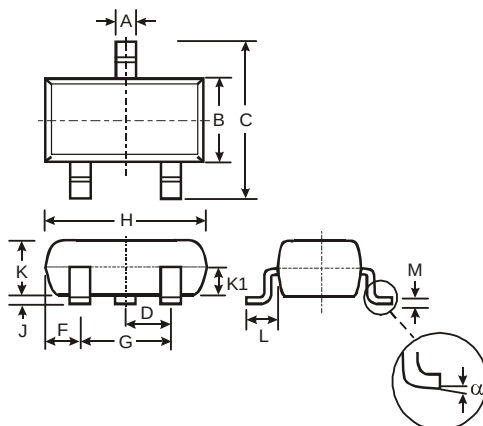
xxx = Product Type Marking Code
(See Electrical Characteristics Table)
YM = Date Code Marking
Y = Year (ex: N = 2002)
M = Month (ex: 9 = September)

Date Code Key

Year	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015
Code	J	K	L	M	N	P	R	S	T	U	V	W	X	Y	Z	A	B	C

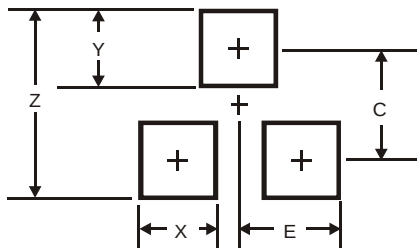
Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Code	1	2	3	4	5	6	7	8	9	O	N	D

Package Outline Dimensions



SOT-23			
Dim	Min	Max	Typ
A	0.37	0.51	0.40
B	1.20	1.40	1.30
C	2.30	2.50	2.40
D	0.89	1.03	0.915
F	0.45	0.60	0.535
G	1.78	2.05	1.83
H	2.80	3.00	2.90
J	0.013	0.10	0.05
K	0.903	1.10	1.00
K1	-	-	0.400
L	0.45	0.61	0.55
M	0.085	0.18	0.11
α	0°	8°	-
All Dimensions in mm			

Suggested Pad Layout



Dimensions	Value (in mm)
Z	2.9
X	0.8
Y	0.9
C	2.0
E	1.35

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